

Corporate Social Responsibility, Competition, and Firm Value

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Abstract

In this paper, we examine the value-enhancing effect of corporate social responsibility in the presence of product market competition. Using a comprehensive sample of 23,505 firm-year observations from 63 countries, we find that the value-enhancing effect of CSR is dominant in non-competitive industries. We investigate whether financial constraints reduce the value of CSR. We show that less financially constrained firms in competitive industries are unlikely to benefit from CSR. Next, we examine the channels through which firms earn positive benefits of CSR. We document a strong positive link between firm profitability, productivity, efficiency, and CSR in non-competitive industries. Overall, results suggest that the value-enhancing effect of CSR is dependent on the product market competition.

JEL Classifications: G10; G11; M14; D40

Keywords: Corporate Social Responsibility; Product Market Competition; Financial Performance; Financial Constraints; Country-Level Determinants

1. Introduction

Researchers have long recognised product market competition as “the most powerful force toward economic efficiency in the world” (Shleifer and Vishny, 1997 p.738). More recently, scholars have focussed on the effect of product market competition on the corporate social responsibility (see Flammer, 2014 among others). These studies show that competition in the industry can influence a firm’s incentive to engage in corporate social responsibility (CSR). However, relatively less is known about the value-enhancing effect of CSR in competitive and non-competitive industries. We show that the value-enhancing effect of CSR is dominant in non-competitive industries.

The growing awareness about environment, social, and governance (ESG) has increasingly put pressure on firms to disclose their commitment towards broad social welfare (Starks, 2009). A 2013 survey undertaken by UN Global Compact-Accenture CEO, covering 1,000 global CEOs from 27 industries and 103 countries, reports that 93 percent of CEOs see sustainability as an important factor in influencing the future success of their business. The increasing drive to undertake CSR activities has led to a rich literature that debates whether undertaking CSR activities can economically and financially benefit a firm. While many studies exist that investigate the impact of CSR on financial performance, there has been relatively less empirical work on the value-enhancing effect of CSR in the presence of product market competition (Kramer and Porter, 2007; Mackey et al., 2007).

Using a comprehensive sample of 23,505 firm-year observations from 63 countries, we find that the positive effect of CSR is dominant in non-competitive industries. The results remain statistically insignificant in the competitive industries. Thus, our results suggest that the benefits of CSR do not equally benefit all firms and product market competition is an important determining factor.

We also investigate whether financial constraints are important factor in explaining the value-enhancing effect of CSR. Using various proxies of financial constraints at the firm-level, we find that less financially constrained firm in competitive industries are unlikely to benefit from CSR. Thus, our results are at odds with Hong et al. (2012) who find that financial constraints are important determinant of CSR.

We then move on to examine the different channels through which some firms are able to earn positive benefits of CSR. We show a strong positive link between firm profitability, productivity, efficiency, and CSR in non-competitive industries. This is consistent with Kitzmüller and Shimshack (2012) who note that CSR improves labour productivity. Finally, we do not find strong evidence of country-level competition laws and scrutiny from public in explaining the dominant effect of CSR. However, the positive impact of CSR on valuation is observed in Civil law origin countries.

This study contributes to the existing literature in a number of ways. First, we show that product market competition is an important determinant in explaining the effect of CSR on financial performance. Although previous studies (see Flammer, 2014 among others) study how product market competition affects the firm's decision to undertake CSR, they do not concentrate on the valuation effect.

Second, by using a comprehensive sample of 23,505 firm-year observations from 63 countries, we show the relationship between competitiveness, CSR, and firm valuation on a global scale. Undertaking a study on a large number of countries sheds new evidence on international markets that are yet to be explored (Kitzmüller and Shimshack, 2012). This is important as majority of the studies study industrialized economies and relatively less is known about developing economies. Therefore, our results address cross-country differences in the investors' perception towards CSR and show whether the results are generalizable or

systematic across countries. We also provide additional evidence on how the country-level determinants influence the firm-level commitment towards CSR and financial performance.

Third, to improve the accuracy of product market competition measure, the study analyses both public and private firms' sales data. The sales Herfindahl-Hirschman Index (HHI) is calculated based on 3,847,684 private and 54,784 publicly listed firms domiciled in 63 countries, from 2004 to 2012. This measure addresses a key problem plaguing the competitiveness measure that uses only public firms' data and ignore the sales figure of private firms.

Fourth, previous studies mostly use KLD dataset as a CSR performance benchmark (Waddock, 2003). In contrast, we build our own measure of CSR from the data provided by Asset4. This CSR index includes 153 individual indicators covering environmental, economic, social, and governance pillars. Thus, we provide evidence on whether the results are consistent if alternative CSR performance benchmark is used.

Finally, we address the important issue of endogeneity between firm valuation and CSR, where it can be argued that financially strong firms are more likely to engage in CSR and therefore causality may run in both directions (Baron, 2008). We use two-stage least squares regression (2SLS), lag CSR scores, system Generalized Method of Moments and introduce an exogenous shock to address these concerns. Our results do not weaken using various tests of endogeneity.

Our study has important policy implications for government policy makers, regulators, and managers. The results from this study suggest that encouraging firms to engage in CSR activities, especially in non-competitive industries, can have a positive influence on firm valuation, productivity, and efficiency. Recently, there has been an increasing emphasis to enforce the accountability of firms towards social, environment, and governance issues. For

instance, the European Parliament and the European Council finalized a deal in 2014 that will mandate large publicly listed firms with at least 500 employees to disclose their environmental, diversity, and social impacts. Our results will help regulators and policy makers to consider industry competitiveness as an additional factor while undertaking important decisions.

2. Hypothesis Development

In this section, we provide a brief overview of the extant literature and develop testable hypothesis. We pursue two competing theories from multidisciplinary literature that support as well as reject the benefits of CSR in a competitive industry. The strategic view of CSR promotes the idea of “do well by doing good” where the firm can be sustainable if stakeholders' interests are taken into consideration (McWilliams and Siegel, 2001). The strategic view suggests that a firm can improve its competitiveness in an industry by undertaking CSR activities. This is possible through multiple channels, such as targeting a niche market, improving the morale and productivity of employees, having access to finance at a lower rate, or to pre-empt costly regulatory actions (Fernández-Kranz and Santaló, 2010). Edward (1933) suggests that in a less than perfect market competition, the firm can create its own loyal customer base by means of product differentiation.

Jones (1995) develops an instrumental theory of stakeholder management and posits that ethical principles (trust, trustworthiness and a cooperative attitude) of a firm can lead to competitive advantage. Flammer (2014) supports this idea and shows that domestic firms increase their CSR activities in response to tariff reductions that are likely to increase competition in the industry. Besley and Ghatak (2005) develop a theoretical model and show that the value of CSR exists in a competitive industry as some consumers are willing to pay a premium for ‘ethical’ products. Therefore, the strategic view of CSR suggests that there will

be a positive correlation between CSR and industry competitiveness (Campbell, 2007). Our first hypothesis is following:

H1: A high CSR in a competitive industry will lead to higher firm valuation.

Contrary views are also expressed in the extant literature on the importance of CSR in non-competitive industries. Davis (1968) develops the idea of ‘iron law of responsibility’ where “those who do not take responsibility for their power ultimately shall lose it.” The author argues that to remain competitive in the long run, a firm cannot ignore social responsibilities; else competitors will step in to assume those responsibilities. Kacperczyk (2009) adds, “with great power comes greater responsibility” and therefore it is critical for a firm to engage in CSR as it contributes to the long-term value. Bagnoli and Watts (2003) present a theoretical model and show that competition decreases the firms’ ability to differentiate product through CSR and therefore the value of CSR will be recognised more in a non-competitive industry.

Baron (2001) provides the intuition behind an increase in social engagement. The author suggests that a government intervention is likely when firms fail to meet the expectations of society, especially in a highly concentrated industry. Similarly, Eesley and Lenox (2006) argue that stakeholders, such as community activists, advocacy groups, religious organizations, and other non-governmental organizations can put pressure on a firm through a set of actions, such as protests, civil suits, and letter-writing campaigns. Therefore, firms in concentrated industries are likely to avoid a confrontation to safeguard their reputation, retain customers, increase employee productivity, and maintain good relations with stakeholders (Kitzmueller and Shimshack, 2012). In addition, Markarian and Santalo’ (2014) and Datta et al. (2013) find that firms operating in concentrated industries are less prone to manipulate earnings, suggesting that a firm can pre-empt regulation by voluntarily adopting practices that

are stringent and stakeholder friendly (Maxwell, et al., 2000). Our second competing hypothesis is as following:

H2: A high CSR in a non-competitive industry will lead to higher firm valuation.

We also investigate whether financial constraints, particularly in competitive industries, can explain the relationship between CSR and firm valuation (Hong et al., 2012). Early scholars, such as Alchian (1950) and Stigler (1958), posit that competition in the product market ensures that management does not waste corporate resources. Schmidt (1997) show that increase in product market competition leads to higher probability of default and reduction in profitability. Therefore, in competitive industries the financial resources allocated to CSR declines as the main concern is to remain competitive. Friedman (1970) and Jensen (2001) add that the main purpose of managers is to run their firms efficiently, rather than to engage in CSR. In a non-competitive industry, however, managers will find it easier to focus on CSR activities due to less financial constraints. Hawn and Kang (2013) find support for this theory as they note that firms in non-competitive industries are less financially constrained, have a wider asset bases, and are, therefore, in a better position to engage in CSR. Allocating resources to CSR in non-competitive industries may be favoured as shareholders prefer spending slack resources on CSR than increasing bonus to the management (Kitzmueller and Shimshack, 2012). Therefore, market competition may act as a disciplinary tool through which the allocation of money to CSR activities is determined. Based on above discussion, our third hypothesis is following:

H3: Financial constrained will benefit less from CSR activities.

3. Data and Methodology

3.1. Corporate Social Responsibility

CSR data comes from Asset4, owned by Thomson Reuters. Asset4 data are categorized into 250 key performance indicators and further into four major categories and 18 sub-categories. The performance of each company is standardized into z-scores that measure the value of units of standard deviation of that value from the mean value of all firms. By the measure of construction, z-scores are normalized to position the score between 0 and 100 percent. However, we do not use this measure, as the exact benchmarks of z-score are not clear. In addition, it is not clear which key performance indicators are used in calculating z-scores and whether they are consistent across time.

To construct our own CSR score from Asset4 dataset, we retrieve information on each economic, environment, social, and governance-related indicator provided by Asset4. For each indicator, Asset4 provides binary information on whether or not the firm meets a key performance indicator. We score one if a company fulfils that indicator or else zero. Next, we add all the scores and divide by the total number of key performance indicators to arrive at the overall corporate responsibility index between 0 and 1. This is in line with the previous literature, where an additive index is constructed from a set of individual indicators (Aggarwal and Goodell, 2009; Bebchuk and Cohen, 2005; Drobetz et al., 2004; Gompers et al., 2003). We carefully scrutinize the data and check that the questions reflect positive CSR attributes. After removing indicators with missing or incomplete information, our index is constructed based on 153 individual indicators. We term it as Corporate Social Responsibility Index (CSR). Due to paucity of space, we summarize the number of indicators under each category and report it in Appendix 1 of this paper. However, the full details of 153 indicators are available from authors on request.

3.2. Product Market Competition

We construct sales HHI as a proxy of product market competition. This measure is used extensively in the extant literature, particularly in the industrial organization literature.¹ It is calculated by squaring the market share of each firm in the industry, country, year and then adding the resulting numbers. It can be represented in formula as:

$$Sales\ HHI_{j,c,t} = \sum_{i=1}^{N_j} s_{i,j,c,t}^2 \quad (1)$$

where $s_{i,j,c,t}$ is the sales market share of firm i in industry j , country c , in year t . The range of this measure lies between zero and one, with a high HHI index suggesting monopoly in the given industry. We use multiple sources to calculate this measure. We start with Worldscope database and obtain sales of all publicly listed firms in the database. However, relying just on the public company sales data may result in poor proxy of product market competition as indicated by Ali et al. (2009). The authors argue that the correlation between Compustat-provided industry concentration and the corresponding US Census measure is 13 percent. Further, we cannot use US Census as this measure is only available for the US market, while our data coverage is global.

To address these shortcomings, we use both public and private data to calculate product market competition. Private data is retrieved from Bureau van Dijk's Orbis database. To improve the quality of the data retrieved from Orbis and to limit the impact of "phantom" firms, Borell et al. (2010) suggest imposing some filters on basic information reported by firms. To this end, we restrict the sample to only those firms that have at least US\$ 100,000 in sales. We retrieve information of 14,293,283 firm-year sales data of 3,847,684 private firms spanning 2004 to 2012. Next, we extract the data of all public listed firms from Worldscope. This results in retrieving 318,292 firm-year sales data of 54,784 public firms.

¹ See Tirole (1988) for a detailed discussion.

Subsequently, we match the databases of Orbis and Worldscope by using the first two digits of the International Securities Identification Number (ISIN) code. We then calculate the sales HHI of at least four firms out of 50 firms in the industry with highest sales every year. This is consistent with the US Census that uses the 50 largest firms within each industry or the largest four-firm concentration ratio. We also consider alternatives where the maximum number of firms in an industry filter is not imposed. Similarly, instead of considering all 50 firms, we concentrate only on the top four sales firms. The results remain qualitatively the same in either case.

3.3. Tobin's Q

Our main dependent variable is Tobin's Q. We use this measure as Tobin's Q captures the valuation effect arising from tangible and intangible asset (Jiao, 2010). This is important as our main variable of interest CSR includes intangible capital, such as reputation and human capital. Following Doidge et al. (2007) and Aggarwal et al. (2010), we compute this measure as the sum of total assets less the book value of equity plus the market value of equity, divided by the total assets.

3.4. Industry Classification and Control Variables

We primarily rely on the industry classification provided by (Fama and French, 1997) FF48 classification in calculating product market competition. To verify the robustness of our results, we also consider the classification provided by the Industry Classification Benchmark (ICB), but the results do not differ.

In order to control the effect of other variables known to affect Tobin's Q, we use illiquidity, standard deviation of stock returns, log of total asset, and blockholding. Following Yermack (1996) and Lang and Stulz (1994), we control for firm size by using logarithm of total assets.

McConnell and Servaes (1990) find a significant curvilinear relation between Tobin's Q and the fraction of common stock owned by corporate insiders. Therefore, we use closely held shares (percent shares closely held).

As Fang et al. (2009) find a positive link between stock liquidity and firm value; we include the illiquidity measure proposed by Amihud and Mendelson (1986). In this measure, illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Next, we use the daily standard deviation of stock returns over the last one year as a measure of risk. We obtain stock prices from Datastream and other control variables from Worldscope. The control variables are winsorized at 1 and 99 percentile to minimize the effect of outliers. Finally, we follow Ince and Porter (2006) and Chui et al. (2010) guidelines and data cleaning procedures to address the data issues noted in Datastream.

4. Results

In Table 1, we provide a brief description of the sample along with country-level determinants. The final sample stands at 23,505 firm-year observations from 63 countries, covering developed, developing, and transitional economies. In the second column, we provide firm-year observations per country. The US, Japan, the UK, and Canada dominate the sample with a large number of firm-year observations. In the third column, we report country-year observations. As expected, the sample distribution is uneven and many countries have a small number of observations.

[Insert Table 1 here]

In Columns four, five, six, and seven, we report Competition Law, Newspaper Circulation, Non-Governmental Organizations (NGOs) per million Population, and Voice & Accountability respectively. Competition Law, drawn from Dyck and Zingales (2004), takes

into account the effect of government policies regarding entry and competition on product market competition across countries. This measure asks whether “competition laws prevent unfair competition in your country?” The higher the score, the more effective the competition law. This data is available only for 37 countries, with the US scoring the highest at 5.96 and Indonesia the lowest at 4.42. The next measure we consider is Newspaper Circulation from Dyck and Zingales (2004). This measure reports the circulation of daily newspaper/population. The notion of including this measure comes from Zingales (2000) as they find that an independent press and a wide circulation can influence public opinion and sanction improper behaviour. The third country level characteristic we include is NGOs per million Population for the year 2000. We retrieve this data from World Resources/EarthTrends Database.² The data is defined as NGOs with offices or members in a particular country divided by its population. This measure will indicate how active NGOs are in addressing social issues. The last country-level determinant reported in Column 7 is Voice & Accountability from the worldwide governance indicator maintained by the World Bank. Voice & Accountability captures perceptions of the extent to which a country’s citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association, and a free media. Denmark ranks first in terms of freedom of expression, scoring 1.634, while Saudi Arabia ranks last at -1.779.

We report cross-country mean, standard deviation, and median of sales HHI in each of the FF48 in Table 2. Personal Services, Business Services, Construction, Wholesale, Retail, and Real Estate (mean HHI range: 0.08 to 0.11) have a high degree of competitiveness, whereas Beer & Liquor, Tobacco Products, Aircrafts, Fabricated Products, and Defense (mean HHI range: 0.34 to 0.49) face low competition. The U.S. Department of Justice considers a market with a HHI of less than 0.10 to be a competitive marketplace; a result of 0.10 to 0.18 to be a

² <http://earthtrends.wri.org>

moderately concentrated marketplace; and a result of 0.18 or greater to be a highly concentrated marketplace.³ As per this criterion, we sort the sales HHI scores on median value and find 20 industries falling under competitive marketplace, 17 industries falling under moderately concentrated marketplace, and the remaining 11 industries under highly concentrated marketplace.

[Insert Table 2 here]

In Table 3, we report the distribution of control variables. The mean and median Tobin's Q is 1.688 and 1.337 respectively. The average CSR index of all firms is 0.40, whereas the mean logarithm of total assets is 15.602. The daily average standard deviation of stock returns is 2.30 percent. The mean and median blockholding (percent shares closely held) is 25.3 percent and 19 percent respectively. Summing up, the control variables reported in Table 3 are within the expected range of values. In unreported results, we take the average and median CSR scores in each HHI tercile. The average CSR score in Low, Medium and High sales HHI is .405, .404 and .438 respectively, while the median CSR score is .392, .386 and .424 respectively. The mean difference between Low and High HHI is .033 and statistically significant at the one percent level. Collectively, this suggests that firms in non-competitive industries engage more in CSR.

[Insert Table 3 here]

4.1. Main Results

We follow the approach of Giroud and Mueller (2011) in designing the regression framework and estimate the effect of product market competition and CSR on Tobin's Q. We perform the following regression:

³ http://www.justice.gov/atr/public/guidelines/horiz_book/15.html

$$Q_{i,j,c,t} = \alpha_i + \alpha_t + \beta'(CSR_{i,j,c,t} \times I_{i,j,c,t}) + \sum \gamma' Controls + \varepsilon_{i,j,c,t} \quad (4)$$

where $Q_{i,j,c,t}$ is Tobin's Q of firm i in industry j , country c , in year t . α_i and α_t are firm- and year-fixed effects to control for unobservable heterogeneity and omitted factor that are related to both Tobin's Q and CSR index. The use of fixed effects is in line with Gormley and Matsa (2014) who note that fixed effect estimators are superior to other measures in controlling for unobserved heterogeneity. We also consider country- and industry-fixed effects instead of firm-fixed effects but we do not find qualitatively different results. Additionally, the use of firm-fixed effects control time-invariant factors that may not be fully captured by the country- and industry-fixed effects. Thus, firm-fixed effects subsume country- and industry-fixed effects. $CSR_{i,j,c,t}$ is the CSR index, while $I_{i,j,c,t}$ is a (3x1) vector of sales HHI dummies. The terciles are formed based on whether the firm is located in the lowest HHI industry, the medium HHI industry or the highest HHI industry. Alternatively, we create (2x1) and (4x1) vector of sales HHI dummies to capture varying degrees of competition. We control for log of asset, blockholding, illiquidity, and standard deviation.⁴ We also include sales HHI dummies to control for direct effects of competition on firm value but do not report in regression tables due to a paucity of space.

We report the main results in Table 4. We first check whether CSR has a positive impact on firm valuation without explicitly controlling for product market competition in Model 1. As expected, the CSR is positively related to firm valuation. The CSR index coefficient is 0.282 and statistically significant at the one percent level. In the set of control variables, logarithm of total asset, illiquidity, and standard deviation are negative and statistically significant at the one percent level, while blockholding is statistically not significant. In Model 2, we introduce the interaction between CSR and sales HHI tercile dummies. The results suggest

⁴ Giroud and Mueller (2011) use firm size, firm age, S&P 500 dummy and Delaware dummy as a set of control variables. We include additional variables as the previous literature suggests that they are related to Tobin's Q. Further, the use of firm-level fixed effects automatically captures time-invariant factors, such as S&P 500 dummy, Delaware, number of business segments etc.

that the effect of CSR on firm valuation is insignificant if the firm is in a competitive industry (0.123, *t-stat* 0.11). Alternatively, the effect of CSR on firm valuation becomes much stronger and statistically significant if the firm is in a non-competitive industry. The interaction term between CSR and sales HHI tercile increases from 0.295 (*t-stat* 2.53) to 0.425 (*t-stat* 4.03) as the competition level changes from medium to low.

In Models 3 and 4, we consider alternative competition dummies to verify that results hold strong even if industry competitiveness is changed from terciles to quantile or quartile. In Model 3, we split the competition dummy between high and low, based on median sales HHI. The results are in line with Model 2, indicating the only CSR*HHI (High) coefficient is positive and statistically significant at the one percent level, while CSR*HHI (Low) is statistically not significant. In Model 4, quartiles are formed based on whether the firm is located in the first quarter (lowest sales HHI), second quarter, third quarter or fourth quarter (highest sales HHI). The absence of CSR in affecting firm valuation in competitive industries indicates that the market is not likely to value the CSR engagement. This is consistent with Liang and Renneboog (2014) and Baumol and Blackman (1991) who note that sacrificing profit for social causes in a competitive industry are not valued.

[Insert Table 4 here]

4.2. Financial Constraints

Next, we investigate whether financial constraints are one of the primary determinants of CSR. We propose five different proxies to determine a firm's financial constraints, namely dividend payout, free cash flow to total assets, cash to total assets, capital expenditure to total assets, and R&D expenses to total assets. We argue that firms with high dividend payout ratio, high free cash flow to total assets, high cash to total assets, high capital expenditure to

total assets, or high research and development expenses to total assets are financially strong and therefore less constrained in commitment towards CSR.

As seen in Table 5, Models 2, 4, 6, 8, and 10 represent firms with potentially low financial constraints, while Models 1, 3, 5, 7, and 9 belong to firms with potentially high financial constraints. We first concentrate on firms split on dividend payout ratio and report it in Models 1 and 2. Firms in a low HHI industry, i.e., firms in competitive industries, are unlikely to add value by engaging in CSR, irrespective of whether financial constraints exist. In contrast, firms in non-competitive industries, i.e., CSR*HHI (Medium) and CSR*HHI (High) variable have positive coefficients and are statistically significant at the five percent level. This suggests that financial constraints are not one of the main CSR determinants. We then split the sample based on free cash flow to total assets and report in Models 3 and 4. The results are qualitatively similar as Models 1 and 2, with CSR*HHI (High) statistically significant at the 10 percent level for firms with low free cash flow to total assets. Next, we consider cash to total assets and capex to total assets in Models 5 to 8, and in general, the results are in line with Models 1 to 4. The results from Models 9 and 10 are similar to other proxies of financial constraints. Summing up, the results from Table 5 suggest that financial constraints are potentially not one of the main factors in explaining the value-enhancing role of CSR under varying degrees of competition.

[Insert Table 5 here]

4.3. Productivity, Efficiency, and Profitability

We next investigate the channels through which CSR impact a firm's valuation. In Table 6, we show the effect of CSR on operating performance and productivity. We include commonly used measure of performance- return on equity and profitability. In Model 1, the dependent variable is return on equity, which is net income divided by the book value of

common equity. In Model 2, the dependent variable is net profit margin, which is net income divided by revenue. In Models 3 and 4 we use the measures of labour productivity and cost efficiency, suggested by Giroud and Mueller (2011). In Model 3, the dependent variable is a logarithm of sales divided by total employees and subtracted by country-specific industry median. The measure is deflated by the consumer price index provided by the World Bank to remove the effect of inflation. In Model 4, the dependent variable is the cost of goods sold divided by sales and subtracted by country-specific industry median.

We first concentrate on Model 1. The results indicate a positive relationship between CSR and return on equity, but only in non-competitive industries. The coefficient CSR*HHI (High) is 0.0769 and statistically significant at the one percent level, while CSR*HHI (Low) coefficient is 0.0438 and statistically insignificant. The results are weaker in Model 2 with CSR*HHI (High) statistically insignificant and CSR*HHI (Medium) is positive but statistically significant only at the 10 percent level. The results from Model 3 supports the view that CSR improves labour productivity as the CSR*HHI (High) coefficient is 0.122 (*t-stat* 4.17) but negative and statistically insignificant in CSR*HHI (Low). The results of Model 4 are inconclusive as all competition and CSR interaction terms are statistically insignificant.

[Insert Table 6 here]

4.4. Country-level Determinants

As our sample is from 63 countries, it is essential to control for cross-country differences in competition laws and other macro determinants. This is necessary as in some countries the competition laws are likely to be biased towards some firms or sectors, such as public undertaking that are regulated or protected by the Government. Thus, country-level laws or regulations may discourage competition. In Table 7, we consider four alternative country-

level determinants: Competition Law; Newspaper Circulation; NGOs per million Population; and Voice and Accountability. Competition Law is related to prevent unfair competition in the country. The higher the score, the more effective is the competition law. The other three measures are proxies of public scrutiny that a firm may face in the event of unethical or unscrupulous practices. Under each of the variables, we split the sample based on median scores.

The results from Table 7 indicate that the effect of CSR on a firm's valuation is dominant in non-competitive industries. In Models 1 and 2 (Competition Law), the CSR*HHI (High) coefficient is 0.288 (*t-stat* 2.05) and 0.681 (*t-stat* 4.02) respectively. Similarly, in Newspaper Circulation Models 3 and 4, the CSR*HHI (High) coefficient is 0.393 (*t-stat* 2.05) and 0.383 (*t-stat* 2.97) respectively. The results are the same when we consider NGOs per million Population and Voice and Accountability. The CSR*HHI (Low) coefficient is statistically not significant in any of the models. In summary, the results suggest that the positive effect of CSR on a firm's valuation in non-competitive industries cannot be fully explained by cross-country differences in competition laws or level of scrutiny from public. In unreported results, we split the sample into Common and Civil law origin countries. Consistent with Botero et al. (2004) and Liang and Renneboog (2014) we find that the value of CSR is reflected more in the Civil law origin countries. This is expected as Civil law is directed more towards protecting the rights of stakeholders (Botero et al., 2004; Liang and Renneboog, 2014).

[Insert Table 7 here]

4.5. Robustness Checks

We undertake a number of robustness checks in Table 8 to reinforce the findings of this study. In the first robustness check, we address the concern that the results are driven by a

poor proxy of CSR. In Model 1 of Table 8, we replace our own CSR index with the standardized CSR index provided by Asset4 (A4CSR). The results indicate that the CSR*HHI is statistically significant only in non-competitive industries. Consistent with our prior findings in Table 4, the CSR*HHI (Medium) and CSR*HHI (High) coefficient is 0.00157 (*t-stat* 3.24) and 0.00186 (*t-stat* 3.60) respectively, while CSR*HHI (Low) coefficient is 0.000777 but statistically not significant.

The second concern that needs to be addressed is unequal sample distribution. As seen in Table 1, this sample is dominated by the US firms. Of the total 23,505 observations, the US firms dominate with 6,634 observations, i.e., 28.22 percent of the full sample. Therefore, it may be the case that the US is driving the results.⁵ To negate this concern, we exclude the US firms from the sample and rerun the regression. The results, reported as Model 2 in Table 8, show that CSR*HHI (Medium) and CSR*HHI (High) are positive and statistically significant at the one percent level, while CSR*HHI (Low) coefficient is statistically insignificant. To further test the impact of uneven sample distribution, we run weighted regression that accounts for a number of observations associated with each country as Model 3 in Table 8. More specifically, following Costa et al. (2013), we use the square root of a number of observations in a weighted multi-factor regression model to control for uneven sample size across countries. The benefit of using square root of a number of observations is that standard error of the mean is a function of the square root of n . The coefficient of CSR*HHI (High) is 0.485 and statistically significant at the one percent level, while CSR*HHI (Low) is statistically insignificant. Models 2 and 3 collectively suggest that results remain qualitatively similar even after controlling for uneven sample distribution across countries.

⁵ Further, we also exclude Australia, Canada, Japan and the United Kingdom and rerun the regression. The results remain qualitatively the same.

Another concern that requires attention is that the dependent variable Tobin's Q is not adjusted for industry-specific variations. For example, Lindenberg and Ross (1981) find significant variation in Tobin's Q across different industries. The authors state that some firms have unique products or factors of production that provide them with competitive advantage and therefore they have high Tobin's Q. Following Bebchuk and Cohen (2005) and La Porta et al. (2002), we construct industry-adjusted Tobin's Q as firm Tobin's Q less country-specific industry median Tobin's Q. We calculate industry median Tobin's Q based on FF48 industry classification in each country and year. The results, reported as Model 4 in Table 8, indicate qualitatively similar findings with CSR positively contributing towards firm valuation in a non-competitive market. The CSR*HHI Index (High) coefficient of 0.476 (*t-stat* 4.68) is similar to Model 2, reported in Table 4.

In Model 5 of Table 8, we consider the potential tension and overlapping between the shareholders and stakeholders interests (Shiller, 2013; Liang and Renneboog, 2014; Krüger, 2015). Consequently, we exclude 41 corporate governance indicators and reconstruct CSR index using 112 indicators, instead of 153 indicators. We term it as CSR excluding CG index. The results suggest that impact of corporate governance in explaining the effect of CSR on firm valuation is minimal. Similar to prior results, the value-enhancing effect of CSR (excluding corporate governance) is dominant in the non-competitive industries.

In Model 6 of Table 8, we investigate potential biased standard errors arising from the correlation of residuals across firms and time. To address this issue, we use two-way clustering by firms and time recommended by Petersen (2009). This procedure controls for the correlation among different firms in the same year and different years in the same firm. We again find that the CSR*HHI coefficient is statistically significant only in the non-competitive industries.

In unreported results, we consider longitudinal hierarchical model that is preferable if the data is organised into levels.⁶ Our sample includes firm-level observations from many industries and countries over years and thus the data can be nested at three levels. The first level is firm where repeated observations belong to same firm. The next level is industry where firms can be grouped into industries. The final level is country as the firms are based in a country. Therefore, hierarchical modelling is superior in capturing interdependencies at various levels. Also, the model takes into account uneven sample distribution across countries. We let the intercept vary at the country-, industry- and firm-level. We find that the results (unreported) are qualitatively same as reported in Table 4.

[Insert Table 8 here]

4.6. *Endogeneity and Identification*

In our final test, we address the important issue of endogeneity and simultaneity between the Tobin's Q and CSR. It can be argued that financially strong firms are likely to engage more in CSR and therefore causality may exist between CSR and firm valuation, i.e., successful firms are more likely to pursue CSR. Alternatively, an omitted factor from the regression could affect both the CSR index and Tobin's Q, leading to a spurious correlation. For example, an omitted factor may be related to CSR and Tobin's Q, and therefore the significant effect of CSR on Tobin's Q is primarily due to an omitted variable. However, the use of firm-fixed effect in the regression model addresses this issue to a large extent as it automatically captures time-invariant unobservable heterogeneity. It is also likely that managers have private information and they will invest in CSR activities only if positive benefits can be realized, i.e., the market rewards a firm in a non-competitive industry. To attenuate these concerns, we undertake two-stage least squares (2SLS) regressions, the

⁶ See Searle *et al.* (1992); McCulloch *et al.* (2001); Verbeke and Molenberghs (2009); and Raudenbush and Bryk (2002) for details.

system Generalized Method of Moments (GMM) proposed by Arellano and Bover (1995) and Blundell and Bond (1998) and introduce an exogenous shock.

In Model 1 of Table 9, we use country-industry mean CSR score as an instrument in the 2SLS regression. The rationale for using country-industry mean CSR score as an instrument comes from the extant literature and used by (Cai et al., 2011; Cheng et al., 2014; El Ghouli et al., 2011; Kim et al., 2014; Lopatta and Kaspereit, 2013). Cai et al. (2011) suggest that industry-mean CSR score fulfil both the relevancy condition and the exclusion restriction. Further the authors note that the industry mean CSR score are close proxy of firm-level CSR and less likely correlated with a firm's Tobin's Q. Consistent with our baseline results, CSR*HHI (Medium) and CSR*HHI (High) coefficient are positive and statistically significant, indicating that endogeneity may not be the primary driver of the results. In Model 2 of Table 9, we use lag CSR score (two years lag) to check the relationship between improvement of CSR score and subsequent increase in firm valuation. This is consistent with Schreck (2011) who note that the lagged values of CSR fulfil the criteria of instrumental relevance and exogeneity. The results again show that the CSR effect on firm valuation exists only in non-competitive industry. In Model 3, we consider GMM model. The advantage of System GMM is that instead of relying on an instrument, it uses its own past realizations as well as independent variables that are not strictly exogenous as instrumental variables. The results, after controlling for endogeneity, indicate that CSR*HHI (Medium) and CSR*HHI (High) coefficient are positive and statistically significant at the one percent level, while CSR*HHI (Low) coefficient is statistically not significant.

In Model 4, we introduce an exogenous shock by including only those firms that Asset4 initiated coverage for the first time. The underlying motivation comes from Orlitzky and Benjamin (2001), Orlitzky et al. (2003), and Chatterji and Toffel (2010) where the authors highlight the importance of rating. More specifically, Orlitzky and Benjamin (2001)

undertake a meta-analysis and find that ratings by a third party (reputation rankings) have the strongest negative correlation with financial risk compared to other measures such as CSR disclosures and social audits. Similarly, Vogel (2005) note that rating agencies forms an important mechanism through which the CSR performance can be compared. Chatterji and Toffel (2010) formally test this proposition by examining the responses of the firm that receive corporate environmental ratings issued by prominent rating agencies for the first time. The authors highlight that “because their fortunes rise and fall with key stakeholder groups, firms have an interest in their ratings” (*p.* 918) and therefore the coverage of the firm (previously not rated) will introduce ‘exogenous shock’ and spur responses from the firm. Following the idea of Chatterji and Toffel (2010) and Cheng et al. (2014), we restrict the sample size to those firms that are rated for the first time by Asset4. Our results reported in Model 4 do not weaken when we introduce the exogenous shocks. Our main results remain qualitatively similar to previously reported results. The CSR*HHI (Low) coefficient remain statistically insignificant but CSR*HHI (Medium) and CSR*HHI (High) remain positive and statistically significant at the one percent level.

In Model 5 of Table 9, we investigate the concern that the sample is not representative. As evident from Table 1, the sample coverage is uneven and firms from developed economies dominate. Therefore, it may be the case that the firms covered by Asset4 have similar characteristics, e.g., large and well-known firms. Stated alternatively, we investigate if the results are biased due to nonrandomly selected sample. We follow two-step procedure suggested by Heckman (1979) to address this concern. Specifically, in the first step we estimate a probit regression in which the dependent variable is a dummy variable taking the value of one if the CSR score is available and regress the control variables. Next, we transform the residuals to inverse Mills ratio and use this ratio in the second step as an additional control variable. The results indicate that the inverse Mills ratio is statistically not

significant. Importantly, we find that our main results are still consistent. Thus, we conclude that self-selection bias is unlikely to affect our main findings.

[Insert Table 9 here]

5. Conclusion

Although significant research is undertaken studying the relationship between product market competition and CSR, relatively less is known about the value-enhancing effect of CSR in the presence of product market competition. Using a comprehensive sample of 23,505 firm-year observations from 63 countries, we find results consistent with prior literature that CSR has a positive effect on firm valuation. However, when we introduce the competition factor, the positive effect of CSR on firm valuation is dominant in non-competitive industries. The positive relationship between CSR and firm valuation in a non-competitive market suggests that firms undertake broad social welfare activities in order to maintain their dominance in the long run and potentially pre-empt costly regulatory actions.

We further investigate the idea that financially constrained firms, particularly in a competitive market, do not benefit from undertaking CSR activities. Using various proxies of financial constraints at the firm-level, we find that less financially constrained firms in competitive industries are unlikely to benefit from engaging in CSR. We next examine the channels through which firms are able to earn positive benefits of CSR. We show that CSR has a positive effect on return on equity, profit margin, and labour productivity in non-competitive industries. This suggests that CSR can improve firm profitability, productivity, and efficiency. The results from this study have important implications for government policymakers, regulators, and firm managers. Our results suggest that the benefits of CSR do not equally benefit all firms and product market competition is an important determining factor.

References

- Aggarwal, R., Erel, I., Stulz, R., Williamson, R., 2010. Differences in governance practices between US and foreign firms: Measurement, causes, and consequences. *Review of Financial Studies* 23: 3131–3169.
- Aggarwal, R., Goodell, J., 2009. Markets and institutions in financial intermediation: National characteristics as determinants. *Journal of Banking & Finance* 33: 1770–1780.
- Alchian, A., 1950. Uncertainty, evolution, and economic theory. *Journal of Political Economy* 58: 211–221.
- Ali, A., Klasa, S., Yeung, E., 2009. The limitations of industry concentration measures constructed with Compustat data: Implications for finance research. *Review of Financial Studies* 22: 3839–3871.
- Allen, F., Carletti, E., Marquez, R., 2014. Stakeholder governance, competition, and firm value. *Review of Finance* 19: 1–32.
- Amihud, Y., Mendelson, H., 1986. Asset pricing and the bid–ask spread. *Journal of Financial Economics* 17: 223–249.
- Arellano, M., Bover, O., 1995. Another look at the instrumental variable estimation of error–components models. *Journal of Econometrics* 68: 29–51.
- Bagnoli, M., Watts, S., 2003. Selling to socially responsible consumers: Competition and the private provision of public goods. *Journal of Economics & Management Strategy* 12: 419–445.
- Balakrishnan, K., Cohen, D.A., 2011. Product market competition and financial accounting misreporting. Working paper.
- Barnett, M., Salomon, R., 2012. Does it pay to be really good? Addressing the shape of the relationship between social and financial performance. *Strategic Management Journal* 33: 1304–1320.
- Baron, D., 2001. Private politics, corporate social responsibility, and integrated strategy. *Journal of Economics & Management Strategy* 10: 7–45.
- Baron, D., 2008. Managerial contracting and corporate social responsibility. *Journal of Public Economics* 92: 268–288.
- Baumol, W.J., Blackman, S., 1991. Perfect markets and easy virtue: Business ethics and the invisible hand. Blackwell Cambridge, MA.
- Bebchuk, L., Cohen, A., 2005. The costs of entrenched boards. *Journal of Financial Economics* 78: 409–433.
- Benoit, J., 1984. Financially constrained entry in a game with incomplete information. *Rand Journal of Economics* 15: 490–499.
- Besley, T., Ghatak, M., 2005. Competition and incentives with motivated agents. *American Economic Review* 95: 616–636.
- Blundell, R., Bond, S., 1998. Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics* 87: 115–143.
- Bolton, P., Scharfstein, D., 1990. A theory of predation based on agency problems in financial contracting. *American Economic Review* 80: 93–106.
- Borell, M., Tykvová, T., Schmitt, C., 2010. How do industry characteristics and persistence shape private equity investments? Centre for European Economic Research (ZEW), Mannheim, Germany.
- Botero, J., Djankov, S., La Porta, R., Lopez–de–Silanes, F., Shleifer, A., 2004. The regulation of labor. *Quarterly Journal of Economics* 119: 1339–1382.
- Brammer, S., Millington, A., 2008. Does it pay to be different? An analysis of the relationship between corporate social and financial performance. *Strategic Management Journal* 29: 1325–1343.
- Cai, Y., Jo, H., Pan, C., 2011. Vice or virtue? The impact of corporate social responsibility on executive compensation. *Journal of Business Ethics* 104, 159–173.
- Campbell, J., 2007. Why would corporations behave in socially responsible ways? An institutional theory of corporate social responsibility. *Academy of Management Review* 32: 946–967.
- Chatterji, A.K., Toffel, M.W., 2010. How firms respond to being rated. *Strategic Management Journal* 31: 917–945.
- Cheng, B., Ioannou, I., Serafeim, G., 2014. Corporate social responsibility and access to finance. *Strategic Management Journal* 35: 1–23.
- Chui, A.C., Titman, S., Wei, K.J., 2010. Individualism and momentum around the world. *Journal of Finance* 65, 361–392.
- Costa, B., Crawford, A., Jakob, K., 2013. Does culture influence IPO underpricing? *Journal of Multinational Financial Management* 23: 113–123.
- Datta, S., Iskandar–Datta, M., Singh, V., 2013. Product market power, industry structure, and corporate earnings management. *Journal of Banking & Finance* 37: 3273–3285.
- Davis, K., 1968. Understanding the social responsibility puzzle. *Business Horizons* 10: 45–50.
- Doidge, C., Karolyi, G., Stulz, R., 2007. Why do countries matter so much for corporate governance? *Journal of Financial Economics* 86: 1–39.

- Drobetz, W., Schillhofer, A., Zimmermann, H., 2004. Corporate governance and expected stock returns: Evidence from Germany. *European Financial Management* 10: 267–293.
- Dyck, A., Zingales, L., 2004. Control premiums and the effectiveness of corporate governance systems. *Journal of Applied Corporate Finance* 16: 51–72.
- Edward, H.C., 1933. *The theory of monopolistic competition*. Harvard University Press, Cambridge, Massachusetts.
- Eesley, C., Lenox, M., 2006. Firm responses to secondary stakeholder action. *Strategic Management Journal* 27: 765–781.
- Eichholtz, P., Kok, N., Quigley, J.M., 2010. Doing well by doing good? Green office buildings. *American Economic Review* 100: 2492–2509.
- El Ghouli, S., Guedhami, O., Kwok, C., Mishra, D., 2011. Does corporate social responsibility affect the cost of capital? *Journal of Banking & Finance* 35: 2388–2406.
- Fama, E., French, K., 1997. Industry costs of equity. *Journal of Financial Economics* 43: 153–193.
- Fang, V., Noe, T., Tice, S., 2009. Stock market liquidity and firm value. *Journal of Financial Economics* 94: 150–169.
- Fernández-Kranz, D., Santaló, J., 2010. When necessity becomes a virtue: The effect of product market competition on corporate social responsibility. *Journal of Economics & Management Strategy* 19: 453–487.
- Flammer, C., 2014. Does product market competition foster corporate social responsibility? Evidence from trade liberalization. *Strategic Management Journal* forthcoming.
- Friedman, M., 1970. The social responsibility of business is to increase its profits. *New York Times Magazine*, pp. 225–230.
- Giroud, X., Mueller, H., 2011. Corporate governance, product market competition, and equity prices. *Journal of Finance* 66: 563–600.
- Gompers, P., Ishii, J., Metrick, A., 2003. Corporate governance and equity prices. *Quarterly Journal of Economics* 118: 107–155.
- Gormley, T.A., Matsa, D.A., 2014. Common errors: How to (and not to) control for unobserved heterogeneity. *Review of Financial Studies* 27: 617–661.
- Graham, J., 2000. How big are the tax benefits of debt? *Journal of Finance* 55: 1901–1941.
- Hawn, O., Kang, H.G., 2013. The market for corporate social responsibility (CSR): How industry structure determines CSR. Available at SSRN 2380641.
- Heckman, J.J., 1979. Sample selection bias as a specification error. *Econometrica* 47, 153–161.
- Hillman, A., Keim, G., 2001. Shareholder value, stakeholder management, and social issues: what's the bottom line? *Strategic Management Journal* 22: 125–139.
- Hong, H., Kubik, J.D., Scheinkman, J.A., 2012. Financial constraints on corporate goodness. National Bureau of Economic Research.
- Ince, O., Porter, R., 2006. Individual equity return data from Thomson Datastream: Handle with care! *Journal of Financial Research* 29, 463–479.
- Jensen, M.C., 2001. Value maximization, stakeholder theory and the corporate objective function. *Journal of Applied Corporate Finance* 14: 8–21.
- Jiao, Y., 2010. Stakeholder welfare and firm value. *Journal of Banking & Finance* 34: 2549–2561.
- Jones, T., 1995. Instrumental stakeholder theory: A synthesis of ethics and economics. *Academy of Management Review* 20, 404–437.
- Kacperczyk, A., 2009. With greater power comes greater responsibility? Takeover protection and corporate attention to stakeholders. *Strategic Management Journal* 30: 261–285.
- Kim, Y., Li, H., Li, S., 2014. Corporate social responsibility and stock price crash risk. *Journal of Banking & Finance* 43: 1–13.
- Kitzmueller, M., Shimshack, J., 2012. Economic perspectives on corporate social responsibility. *Journal of Economic Literature* 50: 51–84.
- Kramer, M., Porter, M., 2007. Strategy and society: The link between competitive advantage and corporate social responsibility. *Harvard Business Review*.
- Krüger, P., 2015. Corporate goodness and shareholder wealth. *Journal of Financial Economics* 115: 304–329.
- La Porta, R., López-de-Silanes, F., Shleifer, A., Vishny, R., 2002. Investor protection and corporate valuation. *Journal of Finance* 57: 1147–1170.
- Lang, L., Stulz, R., 1994. Tobin's q, corporate diversification and firm performance. *Journal of Political Economy* 102: 1248–1280.
- Lesmond, D., Ogden, J., Trzcinka, C., 1999. A new estimate of transaction costs. *Review of Financial Studies* 12: 1113–1141.
- Liang, H., Renneboog, L., 2014. Finance and society: On the foundations of corporate social responsibility, Working paper, Tilburg University.

- Lindenberg, E., Ross, S., 1981. Tobin's q ratio and industrial organization. *Journal of Business* 54: 1–32.
- Lopatta, K., Kaspereit, T., 2013. The world capital markets' perception of sustainability and the impact of the financial crisis. *Journal of Business Ethics* 122: 475–500.
- Mackey, A., Mackey, T., Barney, J., 2007. Corporate social responsibility and firm performance: Investor preferences and corporate strategies. *Academy of Management Review* 32: 817–835.
- Markarian, G., Santalo', J., 2014. Product market competition, information and earnings management. *Journal of Business Finance & Accounting* 41, 572–599.
- Maxwell, J.W., Lyon, T.P., Hackett, S.C., 2000. Self-regulation and social welfare: The political economy of corporate environmentalism. *Journal of Law and Economics* 43: 583–618.
- McConnell, J., Servaes, H., 1990. Additional evidence on equity ownership and corporate value. *Journal of Financial Economics* 27: 595–612.
- McCulloch, C.E., Neuhaus, J.M., 2001. Generalized linear mixed models. Wiley Online Library.
- McWilliams, A., Siegel, D., 2001. Corporate social responsibility: A theory of the firm perspective. *Academy of Management Review* 26, 117–127.
- Orlitzky, M., Benjamin, J., 2001. Corporate social performance and firm risk: A meta-analytic review. *Business & Society* 40: 369–396.
- Orlitzky, M., Schmidt, F., Rynes, S., 2003. Corporate social and financial performance: A meta-analysis. *Organization Studies* 24: 403–441.
- Petersen, M., 2009. Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies* 22, 435–480.
- Porter, M., Kramer, M., 2002. The competitive advantage of corporate philanthropy. *Harvard Business Review* 80: 56–68.
- Raudenbush, S.W., Bryk, A.S., 2002. Hierarchical linear models: Applications and data analysis methods. Sage.
- Schmidt, K., 1997. Managerial incentives and product market competition. *Review of Economic Studies* 64: 191–213.
- Schreck, P., 2011. Reviewing the business case for corporate social responsibility: New evidence and analysis. *Journal of Business Ethics* 103, 167–188.
- Searle, S.R., Casella, G., McCulloch, C.E., 2009. Variance components. John Wiley & Sons.
- Shiller, R.J., 2013. Finance and the good society. Princeton University Press.
- Shleifer, A., Vishny, R.W., 1997. A survey of corporate governance. *Journal of Finance* 52: 737–783.
- Starks, L.T., 2009. EFA keynote speech: “Corporate governance and corporate social responsibility: What do investors care about? What should investors care about?”. *Financial Review* 44: 461–468.
- Stigler, G., 1958. Economies of scale. *Journal of Law and Economics* 1: 54–71.
- Tirole, J., 1988. The theory of industrial organization. MIT press.
- Verbeke, G., Molenberghs, G., 2009. Linear mixed models for longitudinal data. Springer Science & Business Media.
- Vogel, D., 2005. The market for virtue: The potential and limits of corporate social responsibility. Brookings Institution, Washington, DC.
- Waddock, S., Graves, S., 1997. The corporate social performance. *Strategic Management Journal* 8: 303–319.
- Waddock, S., 2003. Myths and realities of social investing. *Organization and Environment*, 16, 369–380.
- Zingales, L., 2000. In search of new foundations. *Journal of Finance* 55: 1623–1653.

Appendix 1: Corporate Social Responsibility Index

Pillar	Category	Number of Indicators
1. Corporate Governance		
The corporate governance pillar measures a company's systems and processes, which ensure that its board members and executives act in the best interests of its long term shareholders.		
	i. Board Functions	23
	ii. Compensation Policy	10
	iii. Shareholder Rights	8
2. Economic		
The economic pillar measures a company's capacity to generate sustainable growth and a high return on investment through the efficient use of all its resources.		
	i. Client Loyalty	6
	ii. Performance	6
	iii. Shareholder Loyalty	6
3. Environment		
The environmental pillar measures a company's impact on living and non-living natural systems, including the air, land and water, as well as complete ecosystems.		
	i. Emission Reduction	16
	ii. Product Innovation	9
	iii. Resource Reduction	13
4. Social		
The social pillar measures a company's capacity to generate trust and loyalty with its workforce, customers and society, through its use of best management practices.		
	i. Community	10
	ii. Diversity and Opportunity	8
	iii. Employment Quality	8
	iv. Human Rights	7
	v. Health & Safety	7
	vi. Product Responsibility	8
	vii. Training & Development	8
Total Number of Indicators		153

Due to paucity of space, we do not list individual indicators. However, the full list is available from the authors on request.
Source: <http://thomsonreuters.com/en/about-us/corporate-responsibility/esg-performance.html>.

Appendix 2: Variable definitions

Variable		Definition	Source
A. Country-level variables			
1	Competition law	Effect of government policies regarding entry and competition on product market competition across countries	Zingales (2004)
2	Newspaper circulation	Circulation of daily newspaper/population	Dyck & Zingales (2004)
3	NGOs per million	NGOs with offices or members in a particular country divided by its population	World Resources/EarthTrends database
4	Voice & Accountability	Perceptions of the extent to which a country's citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association and a free media	World Bank Governance database
B. Firm-level variables			
1	Tobin's Q	Sum of total assets less the book value of equity plus the market value of equity, divided by total assets	Author's own calculation
2	Industry-adjusted Tobin's Q	Tobin's Q less country-specific industry FF48 median Tobin's Q	Author's own calculation
3	Log of asset	Logarithm of total asset	Worldscope
4	Blockholding	Blockholding is percentage of shares closely held	Worldscope
5	Illiquidity	Following Amihud and Mendelson (1986), we calculate illiquidity as the daily ratio of absolute stock return to dollar volume	Author's own calculation
6	Firm risk	Firm risk is volatility of daily returns over the last one year.	Author's own calculation
7	Dividend payout	Dividend payout ratio	Datastream
8	Free cash flow to total assets	Free cash flow to total assets	Datastream
9	Cash to total assets	Cash and short term investment to total assets	Worldscope
10	Capital Expenditure to total assets	Capital Expenditure to total assets	Worldscope
11	R&D expenses to total assets	Research and development expenses to total assets	Worldscope
12	Return on equity	Net income divided by the book value of common equity	Worldscope
13	Net profit margin	Net income divided by sales/revenue	Worldscope
14	Log/sales employee	Logarithm of sales divided by total employees and subtracted by country-specific industry median using FF48 classification in the same country. The measure is deflated by the consumer price index provided by the World Bank to remove the effect of inflation	Author's own calculation
15	COGS/Sales	Cost of goods sold divided by sales and subtracted by country-specific industry median using FF48 classification in the same country	Author's own calculation
C. Product market competition measures			
1	Sales HHI	Sales HHI index is calculated by squaring the market share of each firm in the country-specific FF48 industry and then adding the resulting numbers. Both the public and private firms' data are used	Author's own calculation
D. Industry classification			
1	FF48	Industry classification provided by Fama and French (1997)	Professor French's website

Table 1: Sample distribution

No	Country	Firm-Years	Country-Years	Competition Law	Newspaper Circulation	NGOs per million	Voice & Accountability
1	Australia	1,531	9	5.52	3	196	1.451
2	Austria	133	9	5.29	2.9	529	1.419
3	Belgium	175	9	NA	NA	541	1.375
4	Bermuda	367	9	NA	NA	NA	1.033
5	Brazil	166	8	4.9	0.4	18	0.481
6	Canada	1,548	9	5.37	1.6	133	1.435
7	Cayman Islands	207	9	NA	NA	NA	0.597
8	Chile	71	6	5.4	1	140	1.049
9	China	272	8	NA	NA	2	-1.610
10	Colombia	27	4	4.71	0.5	45	-0.119
11	Cyprus	8	8	NA	NA	NA	1.038
12	Czech Republic	14	6	4.89	2.5	292	0.989
13	Denmark	158	9	5.16	3.1	914	1.634
14	Egypt	16	4	4.6	0.4	28	-0.977
15	Finland	204	9	5.26	4.6	829	1.569
16	France	653	9	5.83	2.2	118	1.282
17	Germany	466	9	5.91	3.1	75	1.374
18	Greece	83	9	NA	NA	335	0.893
19	Hong Kong	346	9	5.85	8	NA	0.528
20	Hungary	15	5	NA	NA	329	0.863
21	Iceland	6	2	NA	NA	5819	1.480
22	India	326	7	NA	NA	3	0.409
23	Indonesia	27	5	4.42	0.2	9	-0.039
24	Ireland	198	9	NA	NA	941	1.416
25	Israel	49	9	5.11	2.9	383	0.603
26	Italy	346	9	5.14	1	98	1.013
27	Japan	3,188	9	5.64	5.8	28	1.004
28	Jersey- British Crown	32	9	NA	NA	NA	1.240
29	Jordan	3	3	NA	NA	133	-0.747
30	Kuwait	6	2	NA	NA	369	-0.567
31	Liberia	2	2	NA	NA	140	-0.805
32	Luxembourg	36	9	NA	NA	NA	1.572
33	Malaysia	145	6	4.84	1.6	83	-0.461
34	Marshall Islands	4	4	NA	NA	NA	1.163
35	Mauritius	5	5	NA	NA	NA	0.818
36	Mexico	42	5	4.93	1	27	0.118
37	Morocco	9	5	NA	NA	47	-0.712
38	Netherlands	248	9	5.53	3.1	392	1.586
39	New Zealand	68	9	5.4	2.2	687	1.573
40	Nigeria	4	4	NA	NA	14	-0.785
41	Norway	144	9	4.96	5.9	918	1.629
42	Oman	4	4	NA	NA	148	-0.995
43	Peru	4	2	5.05	0.8	66	0.080
44	Philippines	58	5	4.61	0.8	26	-0.043
45	Poland	84	6	5.06	1.1	87	1.026
46	Portugal	84	9	4.81	0.8	390	1.192
47	Puerto Rico	5	5	NA	NA	NA	0.842
48	Qatar	5	5	NA	NA	NA	-0.882
49	Russia	125	7	NA	NA	19	-0.891
50	Saudi Arabia	16	6	NA	NA	48	-1.779
51	Singapore	342	9	5.21	3.2	477	-0.165
52	South Africa	344	6	4.89	0.34	67	0.569
53	South Korea	236	7	4.9	3.9	45	0.708
54	Spain	336	9	5.07	1	134	1.132
55	Sri Lanka	3	3	NA	NA	69	-0.550
56	Sweden	389	9	5.08	4.5	559	1.594
57	Switzerland	470	9	5.22	3.3	673	1.612
58	Taiwan	421	9	5.56	2.7	NA	0.818
59	Thailand	79	5	4.77	0.6	29	-0.432
60	Turkey	88	5	5.14	1.1	33	-0.143
61	United Arab Emirates	2	2	NA	NA	295	-0.870
62	United Kingdom	2,428	9	5.74	3.3	128	1.367
63	United States	6,634	9	5.96	2.12	22	1.137
Total (Average)		23,505	428	5.18	2.34	344	0.526

This table shows sample distribution and observations per country. Competition Law is from Dyck & Zingales (2004); Newspaper Circulation is from Dyck & Zingales (2004); NGOs per million population is from World Resources/ EarthTrends Database; and Voice & Accountability is from World Governance Index, maintained by World Bank.

Table 2: Industry-wise sales Herfindahl-Hirschman Index (HHI)

FF48	Description	Mean	StDev	Median	Observations
1	Agriculture	0.31	0.16	0.30	90
2	Food Products	0.16	0.16	0.09	471
3	Candy & Soda	0.31	0.26	0.18	155
4	Beer & Liquor	0.34	0.21	0.32	140
5	Tobacco Products	0.39	0.17	0.29	67
6	Recreation	0.25	0.15	0.20	142
7	Entertainment	0.21	0.18	0.18	135
8	Printing and Publishing	0.18	0.16	0.10	241
9	Consumer Goods	0.22	0.18	0.16	387
10	Apparel	0.20	0.14	0.13	139
11	Healthcare	0.14	0.09	0.09	163
12	Medical Equipment	0.20	0.20	0.08	333
13	Pharmaceutical Products	0.21	0.20	0.10	611
14	Chemicals	0.13	0.11	0.07	744
15	Rubber and Plastic Products	0.12	0.10	0.08	71
16	Textiles	0.18	0.08	0.21	44
17	Construction Materials	0.16	0.17	0.08	478
18	Construction	0.09	0.09	0.05	806
19	Steel Works Etc	0.18	0.16	0.11	508
20	Fabricated Products	0.43	0.34	0.39	13
21	Machinery	0.16	0.12	0.13	770
22	Electrical Equipment	0.22	0.25	0.13	220
23	Automobiles and Trucks	0.19	0.13	0.13	541
24	Aircraft	0.41	0.27	0.25	153
25	Shipbuilding, Railroad Equipment	0.19	0.06	0.18	70
26	Defense	0.49	0.39	0.26	6
27	Precious Metals	0.21	0.13	0.16	388
28	Non-Metallic and Industrial Metal Mining	0.27	0.15	0.24	529
29	Coal	0.19	0.14	0.13	162
30	Petroleum and Natural Gas	0.21	0.19	0.14	1,272
31	Utilities	0.13	0.14	0.09	1,178
32	Communication	0.28	0.18	0.25	1,013
33	Personal Services	0.08	0.10	0.04	136
34	Business Services	0.08	0.10	0.05	1,713
35	Computers	0.19	0.15	0.15	465
36	Electronic Equipment	0.12	0.13	0.05	886
37	Measuring and Control Equipment	0.15	0.13	0.10	228
38	Business Supplies	0.18	0.16	0.10	223
39	Shipping Containers	0.24	0.22	0.15	93
40	Transportation	0.13	0.14	0.07	988
41	Wholesale	0.11	0.10	0.07	647
42	Retail	0.11	0.08	0.09	1,250
43	Restaurants, Hotels, Motels	0.14	0.11	0.11	354
44	Banking	0.16	0.10	0.12	1,481
45	Insurance	0.19	0.18	0.15	987
46	Real Estate	0.11	0.07	0.08	881
47	Trading	0.14	0.13	0.10	1,023
48	Almost Nothing	0.31	0.21	0.24	110
1-48	All	0.16	0.16	0.11	23,505

This table shows industry-wise breakdown of sales Herfindahl-Hirschman Index (HHI), calculated by squaring the market share of each firm in the industry and then adding the resulting numbers. We use Fama & French 48 industry classification.

Table 3: Main variables

	Distribution						
	Mean	StDev	5th	25th	50th	75th	95th
Tobin's Q	1.688	0.999	0.889	1.066	1.337	1.897	3.748
CSR Index	0.404	0.161	0.163	0.281	0.386	0.523	0.686
Log of Asset	15.602	1.567	13.130	14.576	15.506	16.597	18.367
Blockholding	0.253	0.235	0.002	0.032	0.190	0.418	0.709
Illiquidity	2.95E-08	3.41E-07	3.78E-12	4.78E-11	2.81E-10	1.27E-09	3.61E-08
St. Dev of Stock							
Returns	0.023	0.011	0.010	0.015	0.020	0.027	0.044

This table shows the descriptive statistics of the main dependent and control variables used in the multivariate analysis. Following Doidge, et al. (2004) and Aggarwal et al. (2010), we compute Tobin's Q as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year.

Table 4: Effect of CSR on firm valuation in presence of competition

	[1]	[2]	[3]	[4]
CSR Index	0.282*** (2.97)			
CSR* HHI (Low)		0.0123 (0.11)		
CSR* HHI (Medium)		0.295** (2.53)		
CSR* HHI (High)		0.425*** (4.03)		
CSR* HHI (Low)			0.0994 (0.91)	
CSR* HHI (High)			0.387*** (3.92)	
CSR* HHI (1st Qtr)				-0.105 (-0.84)
CSR* HHI (2nd Qtr)				0.294** (2.44)
CSR* HHI (3rd Qtr)				0.443*** (3.93)
CSR* HHI (4th Qtr)				0.367*** (3.22)
Log of Asset	-0.464*** (-17.16)	-0.497*** (-17.30)	-0.496*** (-17.27)	-0.499*** (-17.42)
Blockholding	-0.0213 (-0.53)	-0.00785 (-0.19)	-0.00649 (-0.16)	-0.00776 (-0.19)
Illiquidity	-125256.5*** (-3.50)	-132528.8*** (-3.24)	-132623.1*** (-3.22)	-134096.8*** (-3.21)
Return St. Dev.	-3.351*** (-4.37)	-4.335*** (-5.27)	-4.311*** (-5.23)	-4.297*** (-5.22)
Firm F.E. ?	Yes	Yes	Yes	Yes
Year F.E. ?	Yes	Yes	Yes	Yes
N	25,366	23,106	23,106	23,106
Adj. R ²	19.8%	21.6%	21.6%	21.8%

This table shows the effect of CSR on firm valuation in presence of competition. The dependent variable is Tobin's Q, calculated as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. HHI index is calculated by squaring the market share of each firm in the industry and then adding the resulting numbers. In model 2, HHI tercile dummies are formed based on whether the firm is located in lowest HHI industry, the medium HHI industry, or the highest HHI industry. In model 3 and 4, we create (2 x 1) and (4 x 1) vector of sales HHI dummies to capture varying degree of competition. In model 2, CSR Index is interacted with HHI dummies, HHI dummies and control variables. In model 3 and 4, we interact CSR with HHI dummies that capture varying degree of competition. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors. The model includes firm and year fixed effects.

Table 5: Resource constraint and CSR effect on firm valuation

	Dividend Payout		Free Cash Flow to Total Assets		Cash to Total Assets		Cap. Exp. to Total Assets		R&D Exp. to Total Assets	
	Low	High	Low	High	Low	High	Low	High	Low	High
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
CSR* HHI (Low)	0.0634 (0.32)	0.0784 (0.55)	-0.130 (-1.31)	-0.0625 (-0.29)	0.0470 (0.39)	-0.0419 (-0.18)	-0.0937 (-0.58)	0.133 (0.79)	0.0772 (0.56)	-0.0296 (-0.15)
CSR* HHI (Medium)	0.462** (2.41)	0.326** (2.13)	0.0801 (0.76)	0.421** (2.15)	0.0759 (0.76)	0.476** (1.98)	0.557*** (3.11)	0.142 (0.88)	0.352** (2.49)	0.253 (1.31)
CSR* HHI (High)	0.423** (2.22)	0.411*** (2.99)	0.168* (1.93)	0.698*** (3.92)	0.0325 (0.29)	0.793*** (3.81)	0.599*** (3.66)	0.380*** (2.63)	0.393*** (2.76)	0.462*** (2.85)
Log of Asset	-0.574*** (-13.01)	-0.444*** (-9.43)	-0.291*** (-9.99)	-0.588*** (-14.31)	-0.364*** (-12.36)	-0.631*** (-13.24)	-0.491*** (-10.29)	-0.540*** (-12.76)	-0.485*** (-13.34)	-0.536*** (-11.29)
Blockholding	-0.0619 (-0.93)	0.0466 (0.89)	-0.0227 (-0.55)	0.0226 (0.31)	-0.0281 (-0.66)	0.0276 (0.35)	-0.0558 (-1.04)	-0.00968 (-0.15)	-0.0181 (-0.39)	0.00756 (0.09)
Illiquidity	-204366.1*** (-2.92)	-97940.3** (-2.19)	-106770.7** (-2.46)	-249332.7*** (-2.80)	-82909.3** (-2.21)	-157190.7* (-1.89)	-179742.2*** (-2.58)	-111710.2** (-2.43)	-121410.1*** (-2.97)	-183164.0 (-1.48)
Return St. Dev.	-5.151*** (-4.11)	-3.989*** (-2.88)	-2.372*** (-3.08)	-7.280*** (-4.39)	-5.741*** (-8.14)	-7.975*** (-3.84)	-3.815*** (-4.03)	-4.986*** (-3.33)	-6.284*** (-6.45)	0.0472 (0.03)
Firm F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	11,040	11,141	11,283	11,300	10,436	10,133	11,483	11,456	14,014	9,092
Adj. R ²	22.9%	19.6%	18.0%	25.0%	26.5%	22.9%	21.6%	22.4%	20.6%	23.5%

In this table we test whether scarcity of resources is the primary determinant for the firm in undertaking CSR. We use five different proxies to determine a firm's resource constraint, viz. dividend payout, free cash flow to total assets, cash to total assets, capital expenditure to total assets and R&D expenses to total assets. We split the sample based on the median values. The dependent variable is Tobin's Q, calculated as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. HHI index is calculated by squaring the market share of each firm in the industry and then adding the resulting numbers. HHI tercile dummies are formed based on whether the firm is located in lowest HHI industry, the medium HHI industry, or the highest HHI industry. In all models, CSR Index is interacted with HHI dummies, HHI dummies and control variables. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors. The model includes firm and year fixed effects.

Table 6: CSR, operating performance, productivity

	Return on Equity	Net Profit Margin	Log Sales/ Employee	COGS/ Sales
	[1]	[2]	[3]	[4]
CSR* HHI (Low)	0.0438 (1.56)	-0.00608 (-0.11)	-0.0198 (-0.71)	-0.0109 (-0.59)
CSR* HHI (Medium)	0.0794*** (2.85)	0.0957* (1.82)	0.0660** (2.52)	-0.00268 (-0.14)
CSR* HHI (High)	0.0769*** (2.99)	0.0614 (1.40)	0.122*** (4.17)	0.0183 (0.96)
Log of Asset	-0.00787 (-1.29)	0.0374*** (3.44)	-0.00209 (-0.51)	-0.00244 (-0.60)
Blockholding	-0.0301*** (-2.89)	0.0518* (1.93)	-0.00893 (-1.23)	0.00685 (0.96)
Illiquidity	-4105.7 (-0.26)	907.9 (0.09)	-6159 (-0.95)	5371.5* (1.78)
Return St. Dev.	-4.477*** (-15.49)	-6.732*** (-8.73)	0.134 (0.86)	0.560*** (3.70)
Firm F.E. ?	Yes	Yes	Yes	Yes
Year F.E. ?	Yes	Yes	Yes	Yes
N	22,623	23,068	20,978	20,102
Adj. R ²	10.9%	3.9%	17.4%	1.2%

In this table we show the channels through which CSR can influence than firm valuation. In model 1, the dependent variable is Return on Equity, which is net income divided by the book value of common equity. In model 2, the dependent variable is net profit margin, which is net income divided by revenue. In model 3, the dependent variable is logarithm of sales divided by total employees and subtracted by industry median using Fama & French 48 industry classification in the same country. The measure is deflated by consumer price index provided by the World Bank to remove the effect of inflation. In model 4, the dependent variable is cost of goods sold divided by sales and subtracted by industry median. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. HHI index is calculated by squaring the market share of each firm in the industry and then adding the resulting numbers. HHI tercile dummies are formed based on whether the firm is located in lowest HHI industry, the medium HHI industry, or the highest HHI industry. In all models, CSR Index is interacted with HHI dummies, HHI dummies and control variables. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors. The model includes firm and year fixed effects.

Table 7: Country-level determinants on CSR and firm valuation

	Competition Law		Newspaper circulation		NGOs per million population		Voice & Accountability	
	Low	High	Low	High	Low	High	Low	High
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
CSR* HHI (Low)	0.219 (1.46)	0.00918 (0.05)	-0.00603 (-0.03)	0.00728 (0.05)	0.0683 (0.44)	0.0189 (0.10)	0.0330 (0.22)	0.0375 (0.22)
CSR* HHI (Medium)	0.423*** (3.28)	0.341* (1.82)	0.254 (1.40)	0.527*** (3.65)	0.265* (1.66)	0.471*** (2.79)	0.287** (2.00)	0.336** (2.02)
CSR* HHI (High)	0.288** (2.05)	0.681*** (4.02)	0.393** (2.05)	0.383*** (2.97)	0.424** (2.48)	0.391*** (2.68)	0.258* (1.80)	0.596*** (4.14)
Log of Asset	-0.497*** (-10.26)	-0.528*** (-14.03)	-0.589*** (-13.60)	-0.446*** (-10.90)	-0.579*** (-13.04)	-0.501*** (-12.68)	-0.459*** (-10.52)	-0.529*** (-14.89)
Blockholding	-0.00643 (-0.13)	-0.100 (-1.42)	0.0238 (0.40)	-0.0901 (-1.58)	0.0594 (0.88)	-0.0389 (-0.75)	0.0727 (1.30)	-0.0660 (-1.11)
Illiquidity	-136448.0*** (-2.61)	-97390.5* (-1.88)	-123474.6* (-1.94)	-140351.3*** (-2.87)	14505.5 (0.63)	-153718.0*** (-3.35)	-55062.5 (-1.29)	-157293.4*** (-3.16)
Return St. Dev.	-4.353*** (-2.93)	-2.434** (-2.41)	-4.566*** (-4.10)	-2.816** (-2.17)	-2.409** (-2.37)	-6.817*** (-5.20)	-1.801 (-1.55)	-5.712*** (-4.93)
Firm F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	10,854	10,392	9,962	11,284	10,879	10,557	10,551	12,533
Adj. R ²	21.2%	24.3%	25.0%	22.0%	23.2%	22.4%	21.3%	23.5%

In this table we consider four alternative country-level determinants- competition law; newspaper circulation; NGOs per million population; and voice and accountability. Competition laws ask whether “competition laws prevent unfair completion in your country”, with higher the score, the more effective is the competition law, from Dyck & Zingales (2004). Newspaper circulation reports circulation of daily newspaper/ population, from Dyck & Zingales (2004). NGOs per million population represents how active are Non-Governmental Organizations in addressing social issues, from World Resources/ EarthTrends Database and Voice & Accountability captures perception of the extent to which a country’s citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association, and a free media, from World Governance Index, maintained by World Bank. Under each of the variables, we split the sample based on the median scores. The dependent variable is Tobin’s Q, calculated as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. In all models, CSR Index is interacted with HHI dummies, HHI dummies and control variables. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors. The model includes firm and year fixed effects.

Table 8: Robustness checks

	A4CSR	Excluding US	Weighted Regression	Industry- adjusted Tobin's Q	CSR Excluding CG	Two-way Clustering
	(1)	(2)	(3)	(4)	(5)	(6)
CSR* HHI (Low)	0.000777 (1.36)	0.0168 (0.13)	0.0817 (0.51)	0.0456 (0.41)	0.0440 (0.47)	0.0123 (0.08)
CSR* HHI (Medium)	0.00157*** (3.24)	0.340*** (2.63)	0.327* (1.94)	0.261** (2.33)	0.252*** (2.67)	0.295** (2.10)
CSR* HHI (High)	0.00186*** (3.60)	0.319*** (2.85)	0.485*** (3.17)	0.476*** (4.68)	0.382*** (4.44)	0.425*** (3.10)
Log of Asset	-0.495*** (-17.28)	-0.451*** (-13.19)	-0.571*** (-15.62)	-0.434*** (-16.50)	-0.497*** (-17.30)	-0.497*** (-8.80)
Blockholding	0.0000171 (0.00)	-0.000524 (-0.01)	-0.0329 (-0.53)	-0.0180 (-0.45)	-0.00750 (-0.18)	-0.00785 (-0.12)
Illiquidity	-131624.8*** (-3.22)	-126695.1*** (-3.07)	-160369.3*** (-3.25)	-125544.7*** (-3.46)	-132193.0*** (-3.23)	-132528.8** (-2.40)
St. Dev.	-4.076*** (-4.90)	-4.933*** (-4.38)	-3.085*** (-3.37)	-3.803*** (-5.10)	-4.298*** (-5.22)	-4.335* (-1.84)
Firm F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E. ?	Yes	Yes	Yes	Yes	Yes	Yes
N	23106	16545	23106	23107	23106	23106
Adj. R ²	21.6%	20.8%	24.1%	8.9%	21.6%	78.1%

In this table we undertake a number of robustness tests. In model 1, we replace our own CSR index with the standardized index (A4CSR) provided by Asset4. A4CSR is standardized into z-scores measuring the value of units of standard deviation of that value from the mean of all firms. In model 2, we exclude US domiciled firms. In model 3, we follow Costa et al. (2013) and we use the square root of a number of observations in a weighted multi-factor regression model to control for uneven sample size across countries. In model 4, we replace Tobin's Q with industry-adjusted Tobin's Q. We calculate industry median Tobin's Q based on FF48 industry classification in each country and year. In model 5, we recompute the CSR score after excluding Corporate Governance indicators. In model 6, we use two-way clustering in calculating standard errors. Except model 4, the dependent variable is Tobin's Q, calculated as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. Except model 5, CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. In all models, CSR Index is interacted with HHI dummies, HHI dummies and control variables. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors. The model includes firm and year fixed effects.

Table 9: Endogeneity checks and exogenous shock

	2SLS (1)	Lag CSR Score (2)	GMM (3)	Exogenous Shock (4)	Sample-Selection Bias (5)
CSR* HHI (Low)	0.205 (1.15)	-0.103 (-0.89)	0.353* (1.77)	0.133 (0.48)	0.0967 (0.78)
CSR* HHI (Medium)	0.538*** (3.16)	0.0627 (0.59)	0.526** (2.10)	0.666*** (2.92)	0.319** (2.52)
CSR* HHI (High)	0.613*** (4.08)	0.229** (2.32)	0.717*** (3.28)	0.650*** (2.91)	0.450*** (3.80)
Log of Asset	-0.499*** (-41.26)	-0.529*** (-14.95)	-0.272*** (-6.97)	-0.410*** (-21.36)	-0.553*** (-14.21)
Blockholding	-0.00776 (-0.25)	0.0102 (0.23)	0.0766 (0.93)	0.328*** (4.00)	0.00167 (0.03)
Illiquidity	-132253.7*** (-7.07)	-176793.4*** (-2.68)	-122400.3*** (-2.86)	-68705.3*** (-2.95)	-128802.5*** (-3.13)
St. Dev.	-4.312*** (-7.04)	-4.815*** (-5.70)	-7.172*** (-4.18)	2.808 (1.13)	-5.807*** (-6.26)
Inverse Mills Ratio					-0.047 (0.73)
Firm F.E.?	Yes	Yes	No	No	Yes
Year F.E.?	Yes	Yes	Yes	Yes	Yes
Industry F.E.?	No	No	Yes	Yes	No
Country F.E.?	No	No	Yes	Yes	No
N	23106	16764	23106	3113	20986
Adj. R ²	21.6%	25.2%		34.3%	22.3%

In this table we undertake endogeneity tests. In model 1, we use country-industry mean CSR as an instrument in the 2SLS regression. In model 2 we use lag CSR score (two years). In model 3, we use GMM model and in model 4 we introduce exogenous shock by including only those firms that are covered by Asset4 for the first time. In Model 5, we use inverse Mills Ratio. The dependent variable is Tobin's Q, calculated as the sum of total assets less the book value of equity plus the market value of equity, divided by total assets. CSR index is constructed based on 153 indicators from economic, environment, social and governance pillars provided by Asset4. In all models, CSR Index is interacted with HHI dummies, HHI dummies and control variables. Due to paucity of space, we only report interaction terms along with control variables. Log of Asset is logarithm of total assets. Blockholding is percentage of closely held shares. We calculate illiquidity measure as Amihud and Mendelson (1986), where illiquidity is calculated as the daily ratio of absolute stock return to dollar volume. Standard deviation is calculated as the volatility of returns over the last one year. Firms are classified as per Fama & French 48 industry classification. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors.